

CHEONGJU INTL, Korea, Republic of

WMO: 471280

Lat: 36.717N

Lon: 127.499E

Elev: 191

StdP: 14.59

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.2	12.1	-2.0	5.0	15.6	1.6	6.0	18.6	16.5	28.7	14.8	28.2	3.0	80	0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.0	91.8	78.8	89.8	77.4	87.7	76.3	80.9	88.4	79.5	86.5	78.5	84.9	6.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	151.6	84.8	77.4	143.4	83.0	76.8	140.7	82.5	44.8	88.4	43.3	87.0	42.1	85.9	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.3	13.7	11.7	DB	2.2	94.8	5.1	2.6	-1.5	96.6	-4.5	98.1	-7.4	99.5	-11.1	101.4	(4)
			WB	1.8	82.9	5.3	2.2	-2.0	84.5	-5.1	85.8	-8.1	87.0	-12.0	88.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.6	27.1	32.5	42.4	53.9	64.4	72.8	78.6	79.2	69.8	57.6	44.1	31.4	(6)
(7)		DBStd	18.99	7.39	7.18	6.60	6.44	4.65	3.77	3.92	4.55	5.02	6.33	7.47	7.33	(7)
(8)		HDD50	2278	709	491	249	32	0	0	0	0	14	207	576		(8)
(9)		HDD65	5115	1174	910	701	335	68	3	0	14	241	628	1041		(9)
(10)		CDD50	3957	0	1	13	151	446	683	886	906	595	248	28	0	(10)
(11)		CDD65	1319	0	0	0	3	49	236	421	441	159	10	0	0	(11)
(12)		CDH74	11476	0	0	1	89	589	1799	3695	4143	1074	86	0	0	(12)
(13)		CDH80	3832	0	0	0	8	104	522	1311	1643	239	5	0	0	(13)
(14)	Wind	WSAvg	4.4	4.0	4.5	5.6	5.7	5.1	4.2	4.5	4.3	3.8	3.6	3.8	3.8	(14)
(15)	Precipitation	PrecAvg	49.6	0.8	1.2	1.7	3.0	3.6	6.0	11.3	11.0	6.0	2.2	1.7	1.1	(15)
(16)		PrecMax	72.0	2.4	3.7	4.3	7.6	7.8	14.5	22.9	32.5	15.9	7.8	6.6	2.3	(16)
(17)		PrecMin	29.5	0.0	0.0	0.2	0.6	0.7	0.9	3.8	2.0	0.6	0.1	0.2	0.2	(17)
(18)		PrecStd	11.1	0.6	0.9	1.0	1.6	1.6	3.8	4.7	6.7	4.2	1.9	1.4	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.3	58.8	69.8	80.7	86.1	90.0	95.3	95.4	88.0	80.4	69.8	55.5	(19)
(20)		MCWB	43.5	50.3	53.4	62.0	65.7	72.0	80.6	79.8	75.1	66.0	58.6	47.5		(20)
(21)		2%	DB	46.1	53.5	64.4	76.7	82.7	87.7	91.6	93.2	85.6	76.7	64.6	50.2	(21)
(22)		MCWB	39.6	44.5	49.9	59.8	64.4	70.7	79.4	79.4	72.9	64.3	55.8	44.3		(22)
(23)		5%	DB	42.5	49.6	60.5	73.1	80.4	85.7	89.5	91.2	82.6	73.2	60.9	46.6	(23)
(24)		MCWB	36.9	41.5	48.2	57.3	63.5	69.8	78.2	78.7	71.0	62.4	53.0	41.6		(24)
(25)		10%	DB	39.1	44.8	55.8	68.4	77.2	82.8	87.4	88.1	80.4	69.9	57.3	43.0	(25)
(26)		MCWB	34.4	38.3	45.9	54.7	62.2	69.3	77.5	77.8	70.0	60.4	50.8	39.2		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.7	53.0	57.2	66.8	71.5	79.1	83.0	82.8	78.2	70.2	61.6	49.9	(27)
(28)		MCDWB	48.5	54.6	64.9	74.5	79.6	81.9	89.9	90.5	83.7	75.3	66.5	53.7		(28)
(29)		2%	WB	40.3	45.1	53.2	62.6	68.5	75.3	81.0	81.0	76.0	67.0	57.5	45.2	(29)
(30)		MCDWB	43.8	51.1	61.0	71.1	77.4	81.7	88.1	89.0	81.8	72.9	62.9	48.9		(30)
(31)		5%	WB	37.2	41.7	49.7	59.4	66.5	73.6	79.7	79.8	74.1	63.9	54.5	42.3	(31)
(32)		MCDWB	41.1	47.5	57.6	69.1	75.0	80.4	86.5	87.4	79.3	70.4	59.7	45.6		(32)
(33)		10%	WB	34.8	38.9	46.9	56.8	64.4	72.0	78.6	78.8	72.1	61.8	51.8	39.3	(33)
(34)		MCDWB	38.2	44.0	54.4	66.3	73.3	78.6	84.8	85.8	77.8	67.9	56.5	42.7		(34)
(35)	Mean Daily Temperature Range	MDBR	19.4	20.6	22.9	24.9	23.1	18.9	14.3	15.0	18.7	22.9	20.9	18.7		(35)
(36)		5% DB	MCDBR	21.7	25.6	30.0	31.5	28.3	23.7	17.8	18.4	20.7	25.1	24.9	22.2	(36)
(37)		MCWBR	15.7	17.4	18.1	16.6	13.0	9.9	7.8	7.6	10.1	14.1	16.5	16.8		(37)
(38)		5% WB	MCDBR	18.7	22.8	25.5	25.4	21.8	17.1	15.2	15.9	16.5	20.8	21.5	20.3	(38)
(39)		MCWBR	14.4	16.7	17.1	15.3	12.0	9.0	7.9	7.7	9.3	12.9	15.5	16.2		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.435	0.483	0.569	0.601	0.612	0.680	0.634	0.572	0.497	0.481	0.465	0.431	(40)	
(41)		taud	1.943	1.867	1.710	1.689	1.706	1.630	1.784	1.931	2.073	1.989	1.969	1.975	(41)	
(42)		Ebn at Noon	224	229	222	225	225	210	219	229	240	228	213	215	(42)	
(43)		Edh at Noon	45	55	70	75	75	81	69	58	48	48	43	41	(43)	
(44)	All-Sky Solar Radiation	RadAvg	790	1027	1341	1556	1756	1610	1306	1357	1266	1127	791	704	(44)	
(45)		RadStd	43	82	137	125	154	136	180	195	142	89	90	50	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	-2.48	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (55 neighbors)	N/A	N/A	-1.48	N/A	N/A	N/A	N/A	N/A	+151	+83

Nomenclature: See separate page